PORTFOLIO CASE STUDY

FINANCIAL TRANSACTION ANALYTICS

& Fraud Detection



From 6.36 million simulated transactions to interpretable fraud-detection insights.

PYTHON / PANDAS / SCIKIT-LEARN / POWER BI

FINANCIAL TRANSACTION ANALYTICS

A large dataset. An exceptionally rare event.

PaySim models financial transactions and includes fraud labels for supervised analysis.

TOTAL TRANSACTIONS

6,362,620

11 original variables · 5 transaction types

FRAUDULENT TRANSACTIONS

8,213

6,354,407 legitimate transactions

FRAUD PREVALENCE

0.129%

Only ~1 fraud in every 775 transactions

WHY CLASS IMBALANCE MATTERS

99.871%

of records are legitimate



A model predicting “legitimate” every time would still exceed 99.8% accuracy. Precision, recall and F1-score are therefore more informative.

Source: 01_data_collection; 03_exploratory_data_analysis | PaySim synthetic dataset

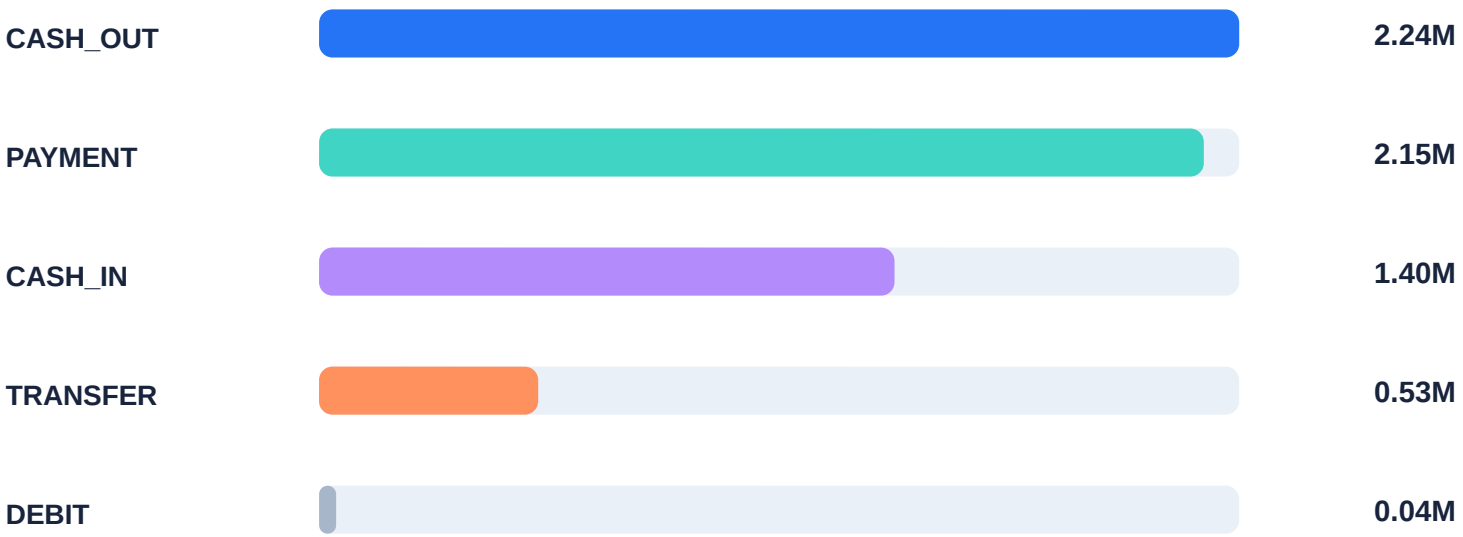
FINANCIAL TRANSACTION ANALYTICS

Transaction activity is concentrated

CASH_OUT and PAYMENT account for most observations; TRANSFER dominates by monetary value.

TRANSACTIONS BY TYPE

Counts (millions)



MONETARY FLOW

42.4%

of total transaction value

is associated with TRANSFER transactions, although they represent only 8.4% of all records.

Monetary values are simulated units (MU), not actual bank balances or losses.

Source: 03_exploratory_data_analysis — volume and value by transaction type

FINANCIAL TRANSACTION ANALYTICS

Fraud concentrates in two transaction types

All 8,213 labeled frauds occur in TRANSFER and CASH_OUT transactions in this dataset.

FRAUD RATE BY TYPE

Share of transactions labeled as fraud



FRAUDULENT VALUE BY TYPE

Share of fraudulent transaction amount



12.06B MU of transaction value associated with fraud

1.0535% of total value vs. 0.1291% of transactions

Source: 03_exploratory_data_analysis | Amounts are not verified financial losses

Models designed for a rare-event problem

Chronological splitting, pre-transaction features and validation-based threshold selection.

01 PREPARE

Validate & encode

No missing values or duplicate records; categorical transaction type one-hot encoded.

02 PARTITION

Split by simulation time

Training: 6,032,860 · Validation: 206,180 · Test: 123,580 transactions.

03 EVALUATE

Choose a threshold

Optimize F1 on validation data; compare precision, recall and errors on test.

MODEL INPUTS

- step
- type
- amount
- oldbalanceOrg
- oldbalanceDest

Logistic Regression

Selected probability threshold: 0.9965

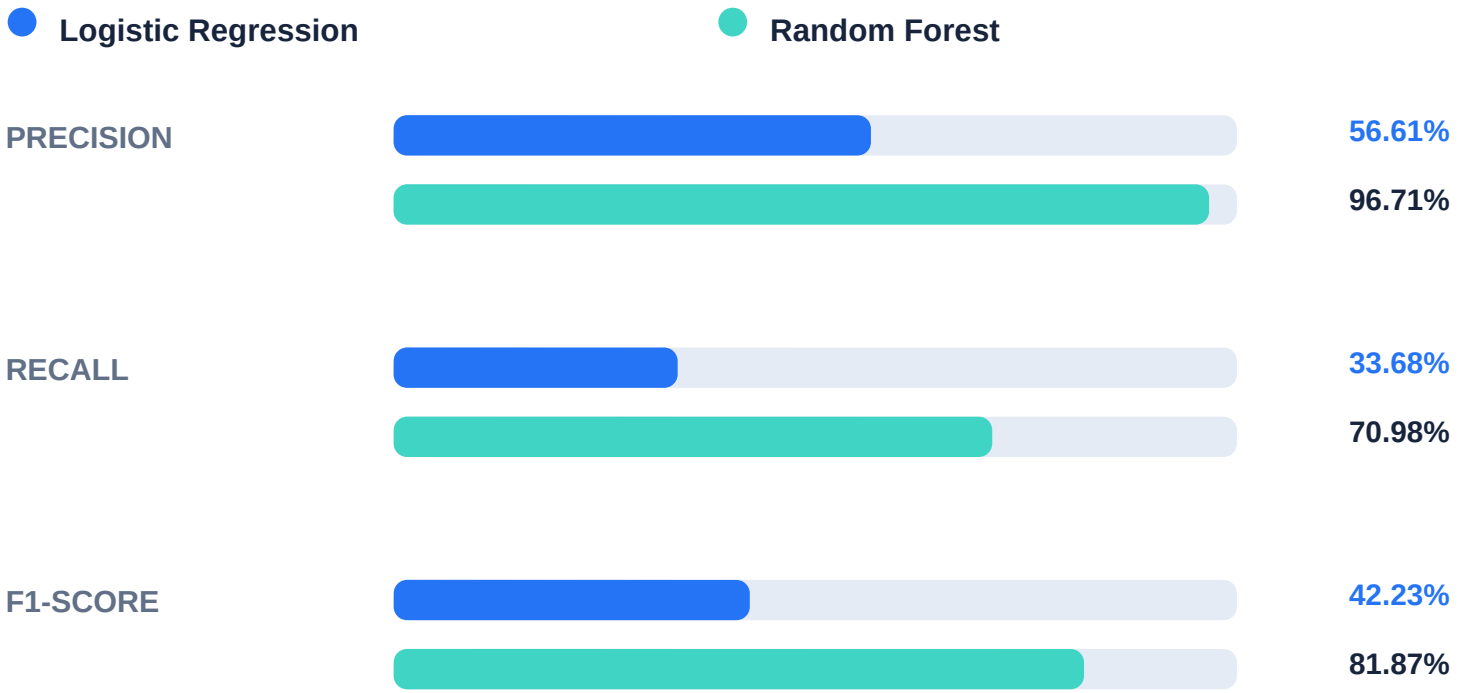
Random Forest

50 trees · max depth 15 · threshold: 0.8701

Source: 02_data_cleaning; 04_fraud_detection — validation and chronological test workflow

Random Forest performs substantially better

Both models were evaluated on the same 123,580-transaction test period (1,654 labeled frauds).



TEST OUTCOMES

1,174

frauds detected by Random Forest

RF false negatives	480
RF false positives	40
LR frauds detected	557

Source: 04_fraud_detection — final test predictions | Metrics refer to fraud class

THE TAKEAWAY

A stronger model, not a finished risk system.

96.71%

Random Forest precision

70.98%

Random Forest recall

480

Frauds still missed

WHAT COMES NEXT

Evaluate financial costs of errors · Validate on a truly untouched test period · Test robustness on external data

Limitations: PaySim is synthetic; the test period was inspected during preliminary experiments. Reported results are exploratory, not production guarantees.